

ELECTRIC VEHICLES · BATTERY SYSTEMS · CONTROL ENGINEERING

Engineering Smarter Battery Management for Electric Vehicles.

I'm Dannana Tharun, an Electronics and Instrumentation Engineering graduate focused on battery pack modeling, battery management systems, active cell balancing, SOC/SOH estimation, and model-based control using MATLAB, Simulink, and Simscape.

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B.Tech, E&I Engineering Adikavi Nannaya University	CGPA 8.2 / 10 2022–2026	IIFE Visakhapatnam Battery systems internship
Active cell balancing Li-ion battery project	MATLAB / Simulink Simscape modeling	

About

My Electronics and Instrumentation background connects measurement, sensing, control systems, simulation and battery management. I am an early-career engineer aiming for entry-level roles as a Battery Management System Engineer, working towards safe, reliable, efficient and intelligent battery systems for EVs.

Practiced in projects
Battery pack modeling, SOC monitoring, active balancing concepts,

Studied
SOH estimation, thermal management, NMC and LFP chemistries, EV battery monitoring

07/10/2026, 05:40

Dannana Tharun – BMS & EV Portfolio

MATLAB/Simulink/Simscape.

and protection.

Still developing

Hardware BMS implementation, embedded firmware, CAN-based communication.

Technical Skills

Grouped by evidence level; no arbitrary proficiency percentages.

Battery & EV

BMS fundamentals · Li-ion systems · pack modeling · SOC/SOH concepts · active cell balancing · thermal management concepts · NMC/LFP awareness

Modeling & Control

MATLAB · Simulink · Simscape · Stateflow · PID/PI control concepts · closed-loop control · PWM concepts

Instrumentation & Automation

Sensors & transmitters · calibration tools · loop troubleshooting · PLC programming · P&ID design

Programming & Protocols

Python fundamentals · basic CAN, I²C, SPI awareness · embedded systems fundamentals

Featured Project

Active Cell Balancing of Lithium-Ion Batteries for an EV

IIPE Visakhapatnam · 8 Dec 2025 – 7 Apr 2026 · B.Tech project report. Guide: Dr. D. Kishore, Assistant Professor, Dept. of E&I Engineering. Teammate: K. Devi Prasad.

Overview

Cells in a pack drift apart in State of Charge (SOC). Without balancing, the weakest or most-discharged cell limits usable pack capacity and consistency. Active balancing moves energy between cells instead of dissipating it.

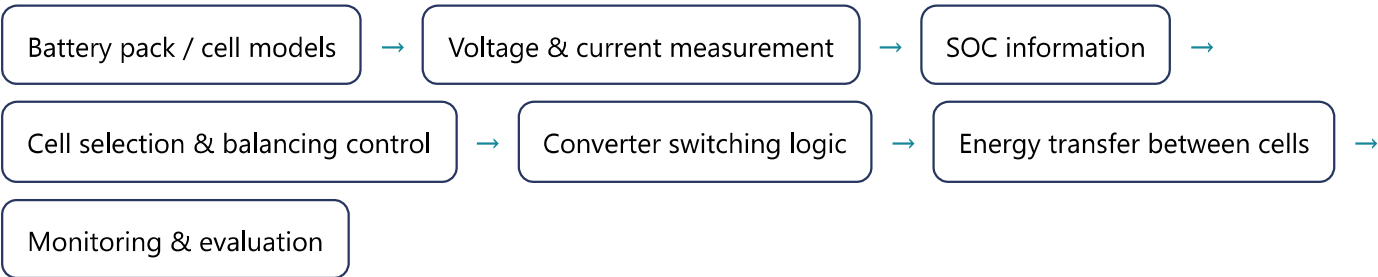
Objectives

Investigate active cell balancing, BMS concepts, SOC/SOH estimation and energy transfer between cells.

Technical approach

MATLAB/Simulink battery modeling and SOC monitoring. The report discusses switched-capacitor and inductor-based DC–DC balancing, buck–boost converter concepts, closed-loop control, PI, PWM and switching logic. These are methods studied and modeled in simulation, not a hardware-ready integrated system.

System architecture (conceptual representation)



Results and engineering analysis

Distribution comparison

Initial SOC values span roughly 40–90%. The

Time-based simulation

The SOC-imbalance curve stays nearly

plotted final distribution spans roughly 65–75%, which suggests the intended balancing outcome.

constant at about a 50% spread, and individual SOC traces change little over the 60-minute run. Expected dynamic balancing is not consistently demonstrated.

Engineering Challenges & Next Steps

The balancing algorithm is not claimed as fully validated. The report points to these areas for further investigation:

- SOC update equations
- Converter current and battery model interaction
- Energy-transfer implementation
- Dynamic simulation and validation

Learning: control logic, battery state updates, converter behavior and measurable simulation results must be connected end to end, otherwise a plausible-looking result can hide a model gap.

Evidence gallery

Simulink model screenshot
(add real image)

Converter subsystem
(add real image)

SOC plots
(add real image)

Cell-selection & control logic
(add real image)

View Project Report

View Completion Certificate

View Technical Figures

Other project & presentation

Conference

Design and Analysis of EMG Signals Using Virtual Instrumentation

ICABMSHT 2024, PSG College of Technology, Coimbatore (Dept. of Biomedical Engineering). EMG signal analysis with virtual instrumentation.

Workshop participation (separate from the paper): “Innovative Technologies in Human Motion”, 19 Dec 2024, with Delsys Inc.

Presentation record

Workshop certificate

Internship Experience

12 May – 12 Jul 2025

Design of Battery Pack and BMS for Electric Vehicles

Indian Institute of Petroleum and Energy, Visakhapatnam. Per the certificate: MATLAB, battery design and simulation, battery-pack modeling, and BMS simulation for EVs using Simscape.

View Certificate

Short-term · May 2024

Lead Resistance Compensation in Resistive Sensors

National Institute of Technology, Tiruchirappalli. Studied lead-resistance compensation techniques to reduce remote measurement errors in resistive sensor applications. Exact dates to be confirmed from the certificate.

View Certificate

Certificates & Training

Design of Battery Pack and BMS for EVs

IIPE Visakhapatnam · 12 May – 12 Jul 2025

[View Certificate](#)

Active Cell Balancing of Lithium-Ion Batteries for an EV

IIPE Visakhapatnam · 8 Dec 2025 – 7 Apr 2026

[View Certificate](#)

One-Day Awareness Program on EV BMS, Practical Demonstration & Expo (EV&BMS-26)

NIT Warangal · 9 Feb 2026 · Hioki and Vastek branding on certificate

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Innovative Technologies in Human Motion (workshop participation)

PSG College of Technology with Delsys Inc. · 19 Dec 2024

[View Certificate](#)

Python for Beginners

Simplilearn SkillUp · 8 Jan 2024 · Code 4769405

[View Certificate](#)

NIT Tiruchirappalli internship

Exact title and date to be added from the certificate file.

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The Coursera “Python for Everybody” credential is listed on the resume but not shown here until its certificate is verified.

Education

2022–2026

2020–2022

B.Tech, Electronics & Instrumentation Engineering

University College of Engineering, Adikavi
Nannaya University · CGPA 8.2/10

Intermediate (MPC)

AP Model Junior College · GPA 7.7/10

2019–2020

SSC

ZPHS · GPA 9.8/10

Career Focus

Roles I am interested in, not positions I hold:

- Battery Management System Engineer
- Battery Modeling and Simulation Engineer
- Battery Testing and Validation Engineer
- EV Systems and Controls Engineer
- Entry-level battery systems and embedded BMS roles

Contact

Recruiters, technical teams and collaborators are welcome to reach out about battery modeling, BMS development and EV engineering opportunities.

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